

## CLINICAL DECISION SUPPORT &amp; MULTI-OMICS DOSSIER

Patient Submitter ID: TCGA-BC-A108 | Tumor: Liver Hepatocellular Carcinoma (LIHC) | Stage: Stage I

|                 |                   |                    |                              |
|-----------------|-------------------|--------------------|------------------------------|
| TCGA Project:   | TCGA-LIHC         | cBioPortal Study:  | lihc_tcga_pan_can_atlas_2018 |
| Gender / Age:   | Female / 58 yrs   | Vital Status / OS: | Alive (44.1 mo)              |
| Hybrid ML Risk: | 74.1% (HIGH RISK) | VQC / Classical:   | 71.0% / 76.2%                |

## FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

| Gene | Locus (GRCh38) | Transcript      | Protein Change               | ClinVar Status             | COSMIC ID                               | gnomAD Pop AF | VAF % |
|------|----------------|-----------------|------------------------------|----------------------------|-----------------------------------------|---------------|-------|
| TP53 | chr17:17p13.1  | ENST00000269305 | p.R249S<br>Missense Mutation | Pathogenic<br>VCV000012384 | COSV52668586<br>FATHMM: 0.99 Pathogenic | 0.000001      | 37.8% |

QUANTUM VARIATIONAL ROTATION STATE ANGLES ( $\theta_{VQC}$  /  $R_Y$  / CNOT)

| Gene Symbol | Canonical Ensembl ID | VAF Hotspot Frequency | Variational Angle $\theta$ (Radians)               | Qubit Hilbert Space Projection                               |
|-------------|----------------------|-----------------------|----------------------------------------------------|--------------------------------------------------------------|
| TP53        | ENSG00000141510      | 37.8%                 | $\theta = 1.9420 \text{ rad (111.3}^\circ\text{)}$ | $ \psi\rangle = \cos(0.971) 0\rangle + \sin(0.971) 1\rangle$ |

## PRECISION ONCOLOGY CLINICAL ACTION &amp; PROTOCOL RECOMMENDATION

## Recommended Therapeutic Pathway:

Surgical segmentectomy resection followed by atezolizumab plus bevacizumab immunotherapy surveillance and CT dynamic liver monitoring.

**Genomic Biomarker Summary:** Identified somatic hotspot mutations in TP53 (p.R249S). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

**NOTICE & RESEARCH DISCLAIMER:** This report is generated by the BlazeFinix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.